Ishita Bhargava

Human-Centred AI ▪ Intelligent System Design ▪ Artificial Intelligence

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EDUCATION

WELLESLEY COLLEGE Wellesley, MA

Bachelor of Arts in Computer Science; Magna Cum Laude *Sep 2023 – May 2026*

- **GPA**: 3.85/4.0, **Major GPA**: 3.93/4.0
- Phi Beta Kappa - Academic Honor Society, Member (2026)
- **Relevant Computer Science Coursework**: Data Structures, Algorithms, Theory of Computation, Foundations of Computer Systems, Natural Language Processing, Human-Computer Interaction, Extended Reality, Advanced Projects in Interactive Media, Combinatorics & Graph Theory, Linear Algebra, Fundamentals of Statistics
- **Certifications**: Generative AI with LLM, Google UX Design Professional Certificate, Full Stack Development Course

MASSACHUSSETTS INSTITUTE OF TECHNOLOGY (MIT) Cambridge, MA

Cross-Registered Student *Feb 2024 – May 2026*

- ***GPA:*** 4.0/4.0; **Relevant Coursework**: Intro to Python, Tangible Interfaces, Design Thinking and Innovation Leadership

SKILLS AND INTERESTS

Computer Languages: Python, JavaScript, TypeScript, SQL, C, C++, C#, Java, HTML/CSS, Assembly
Technical: Git, React, PyTorch, NumPy, Unity, Figma, LLMs, NLP, LangChain, UX Research, Physical Computing, FastAPI
Interests: Photography, Writing, Travelling, Graphic Design, Badminton, Cricket, Theatre

WORK EXPERIENCE

Research Assistant Bangalore, India

AI & Robotics Technology Park (ARTPARK), Indian Institute of Science (IISc) *July 2026 – Present*

- Collected and processed multimodal LiDAR, RGB, fisheye, and IMU data from robotic scans of dynamic construction environments to support persistent 3D digital-twin reconstruction.
- Developed structured scene-graph representations linking construction-site spaces, objects, and elements through spatial and functional relationships for downstream querying and reasoning.

Undergraduate Researcher Wellesley, MA

Wellesley HCI Lab *Sep 2025 – May 2026*

- Developed AI-assisted healthcare interfaces for non-emergency medical decision support.
- Designed and implemented an eye-tracking calibration pipeline using **WebGazer** to support user studies.
- Conducted research on a benchmark for evaluating Multimodal Large Language Models' basic spatial reasoning capabilities.

Undergraduate Research Intern Cambridge, MA

Space Enabled Group - MIT Media Lab *June 2025 – Aug 2025*

- Prototyped and validated interactive simulation systems for the Zero Robotics ISS competition inviting over 200 participants.
- Collaborated with NASA researchers to integrate real-world robotics programming into an educational simulation platform
- Supported global student teams with structured debugging sessions & technical forums, improving system reliability & usability

UI/UX and Game Development Intern Los Angeles, CA

Alterea Inc. (Agents Of Influence – an educational video game) *June 2025 – Aug 2025*

- Developed AI-augmented performance feedback system across 10+ conversational learning modules
- Built interactive, filterable dashboard, visualizing real-time data across conversational analytical game modules
- Improved clarity and usefulness of feedback loops in educational AI system, supporting engagement & learning outcomes

PROJECTS

HOW SPATIALLY INTELLIGENT IS AI? (Published at ACM C&C '26) *Link to Paper* *Sep 2025 – Dec 2025*

- Led an independent study investigating the extent of spatial intelligence in conversational AI through hands-on and benchmark-based experiments.
- Analyzed the AI's spatial instructions and visual reasoning, identifying frequent errors, inconsistencies, and missing embodied understanding.

TANGIBLE INTERFACES: BETWEEN TIME'S FINGERS *Link to Paper* *Sep 2024 – Dec 2024*

- Designed an interactive installation that invites participants to share personal stories about a loved one, fostering a sense of community through shared experiences.
- Created uniformly black, 3D-printed objects for each participant, serving as tangible representations of their stories within the communal display space which used haptic sensors to tell a story.

LEADERSHIP EXPERIENCE

GORDON-MIT ENGINEERING LEADERSHIP PROGRAM Cambridge, MA

Gordon Engineering Leader *June 2025 – May 2026*

- Participating in selective leader development program focused on being an effective leader of industry engineering teams
- Actively practicing leadership, teamwork & communication skills in engineering context; complementing MIT's technical coursework